



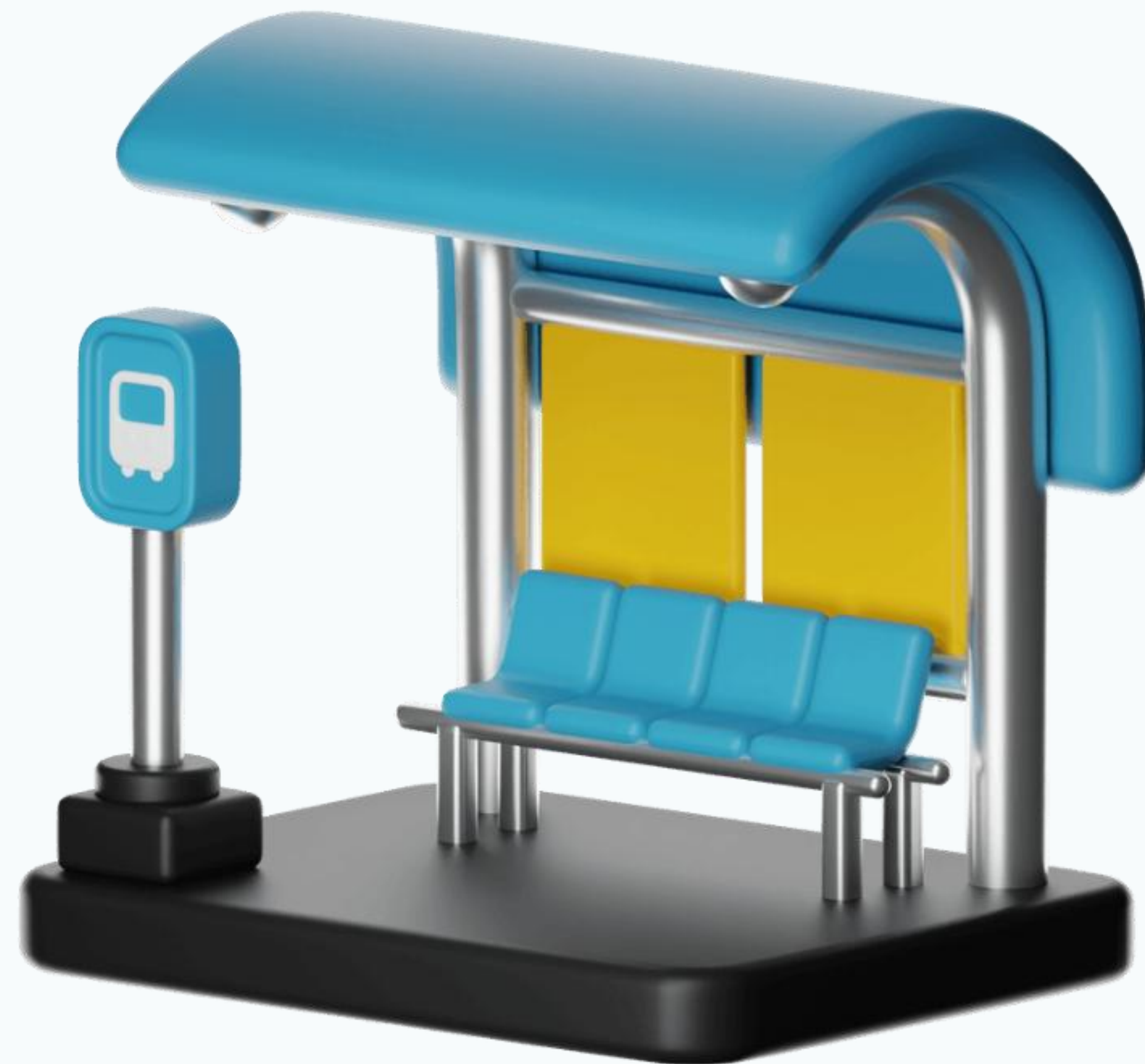
# SoundSync: Smart Transit Application

for Improving Reliability & Connectivity  
in Seattle and Bellevue's public transportation

Team 6

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# 01. Background

## King County Council Wants Fewer 'Ghost' Buses and Better Data Collection

By Stephen Fesler - March 19, 2025

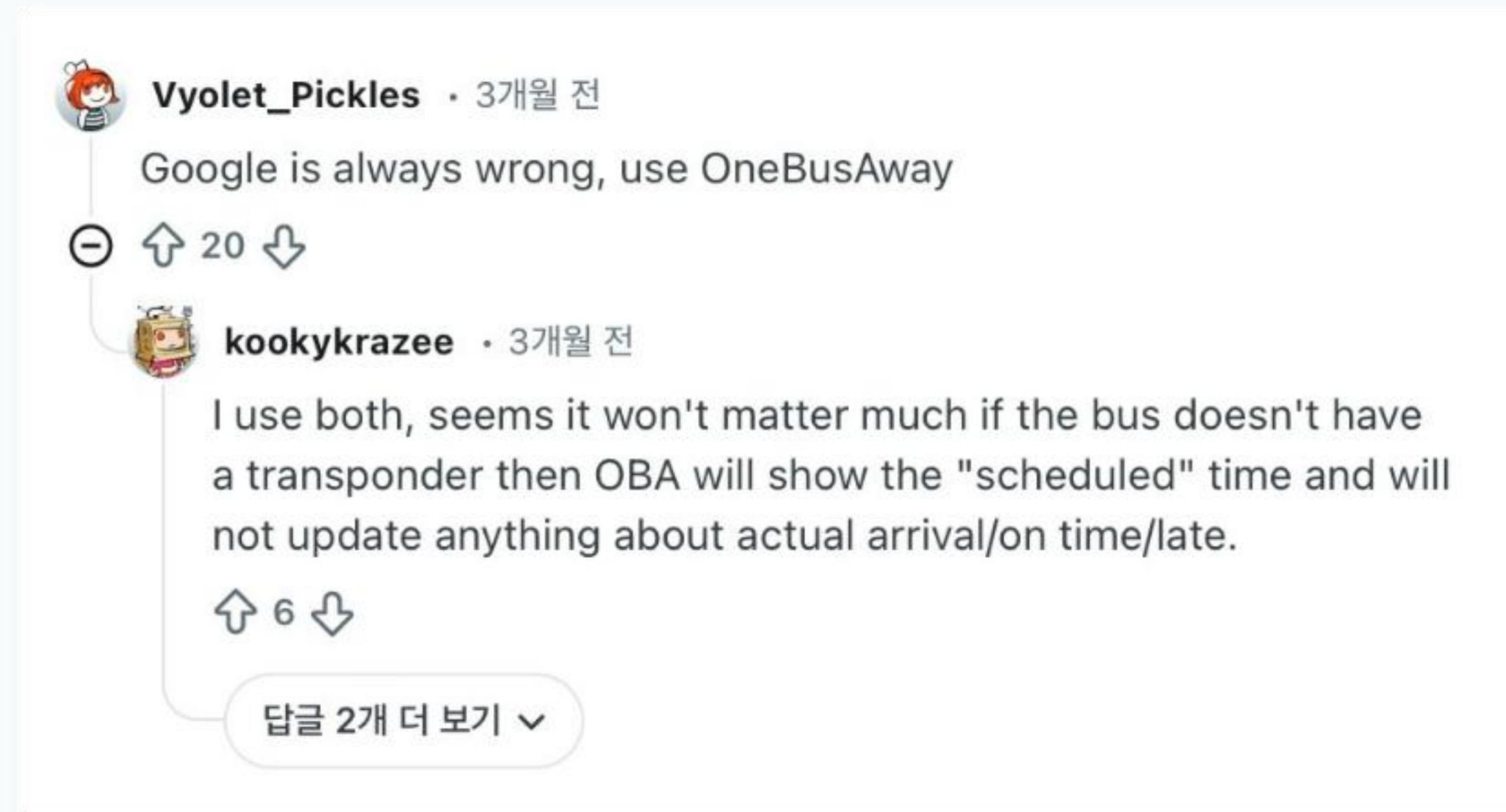
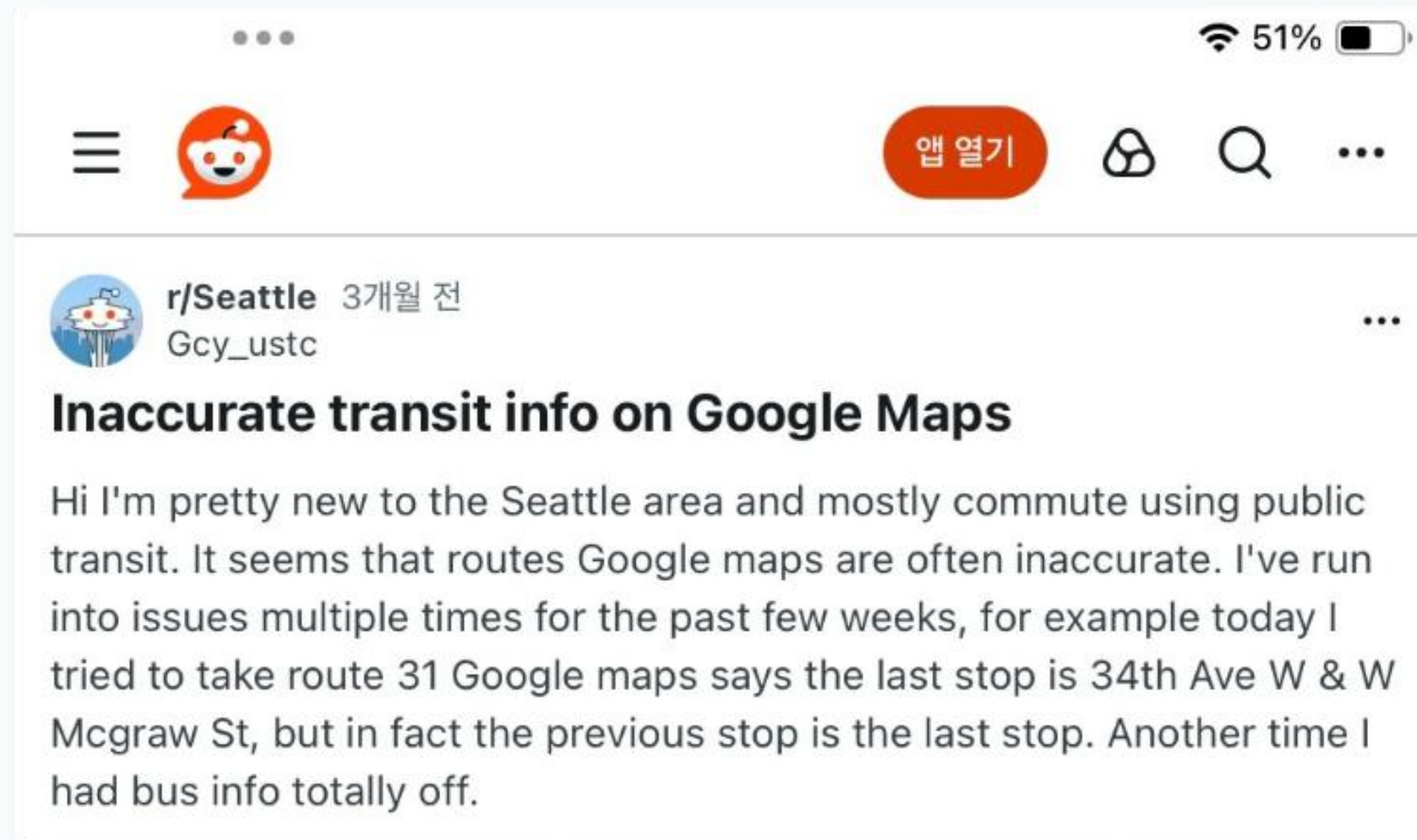
The King County Council wants to zap the frequency of "ghost" buses, which are the bane of many transit riders' existence. A ghost bus is when buses that are scheduled, and even appear inbound on transit apps, never actually come.



Our team actually toured Seattle all day last Sunday and actually looked at the scene of the bus delay and waiting people.



# 01. Background



A new commuter moving to Seattle is **suffering from public transportation**. The main problem is that the schedule is mainly presented on Google Maps and the **exact expected arrival time** is not displayed. Even the **delay time** is not reflected properly.



## 02. Teammate Interview



The bus app clearly indicates that the bus is scheduled to arrive, but often, the bus doesn't actually show up. In those cases, I have no choice but to wait for the next bus, and I eventually end up being late for school."



Bus operations often change due to weather changes or traffic controls, but it is difficult to check all this information at once. I use Google Maps, but since the actual bus status, weather, and traffic information are separated, it is inconvenient to make decisions while on the move.





# 03. Pain Points

## point 1

Information such as bus arrival time, stop location, safety, weather information is fragmented



## point 2

Absence of notification of unexpected situations such as bus delays, disasters, etc





# 04. Main Idea



## SoundSync

Real-Time Transit for Bellevue & Seattle

Real-time Integrated Bus Information System

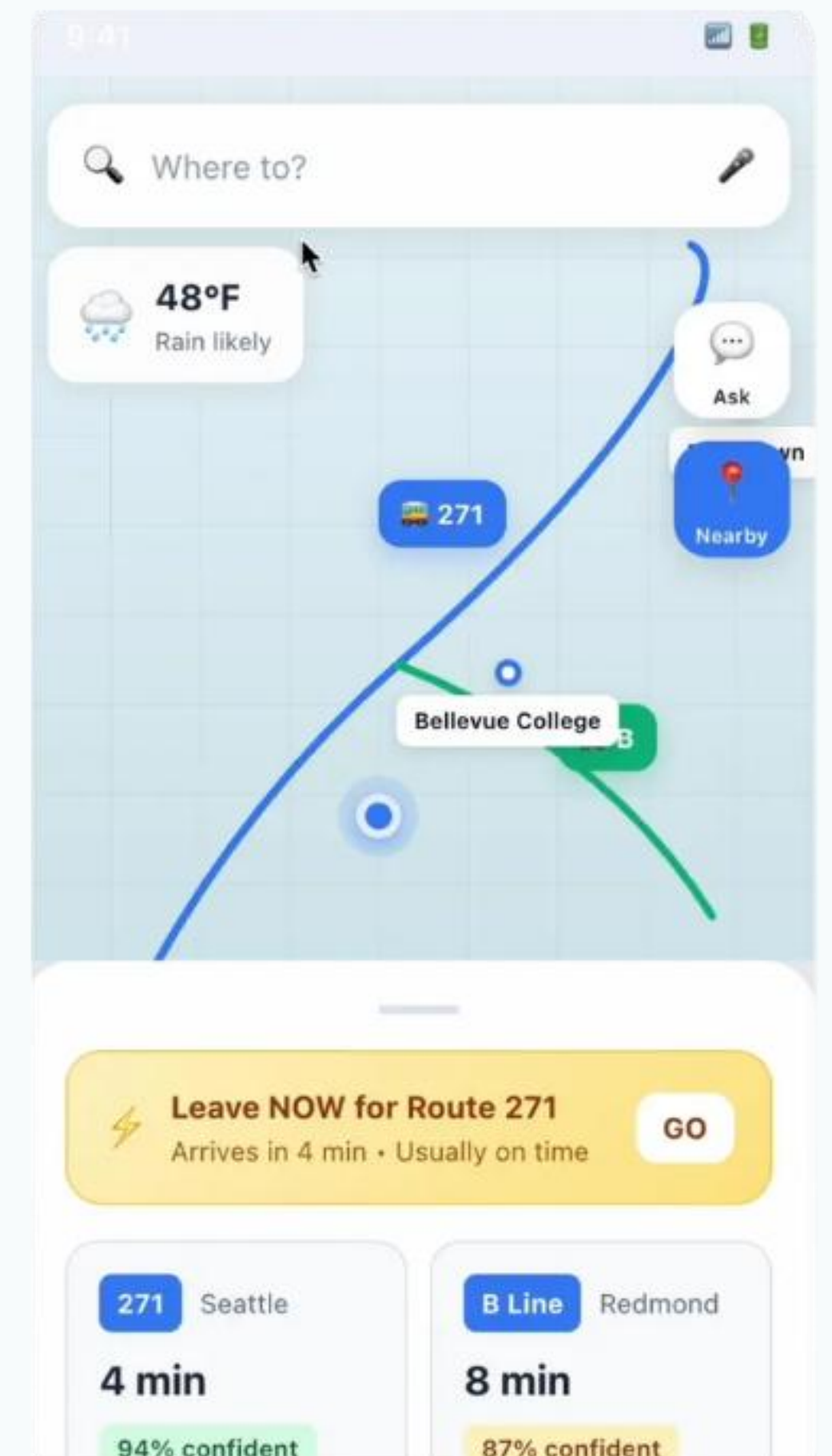
Bus arrival information

Additional integrated data

Weather information

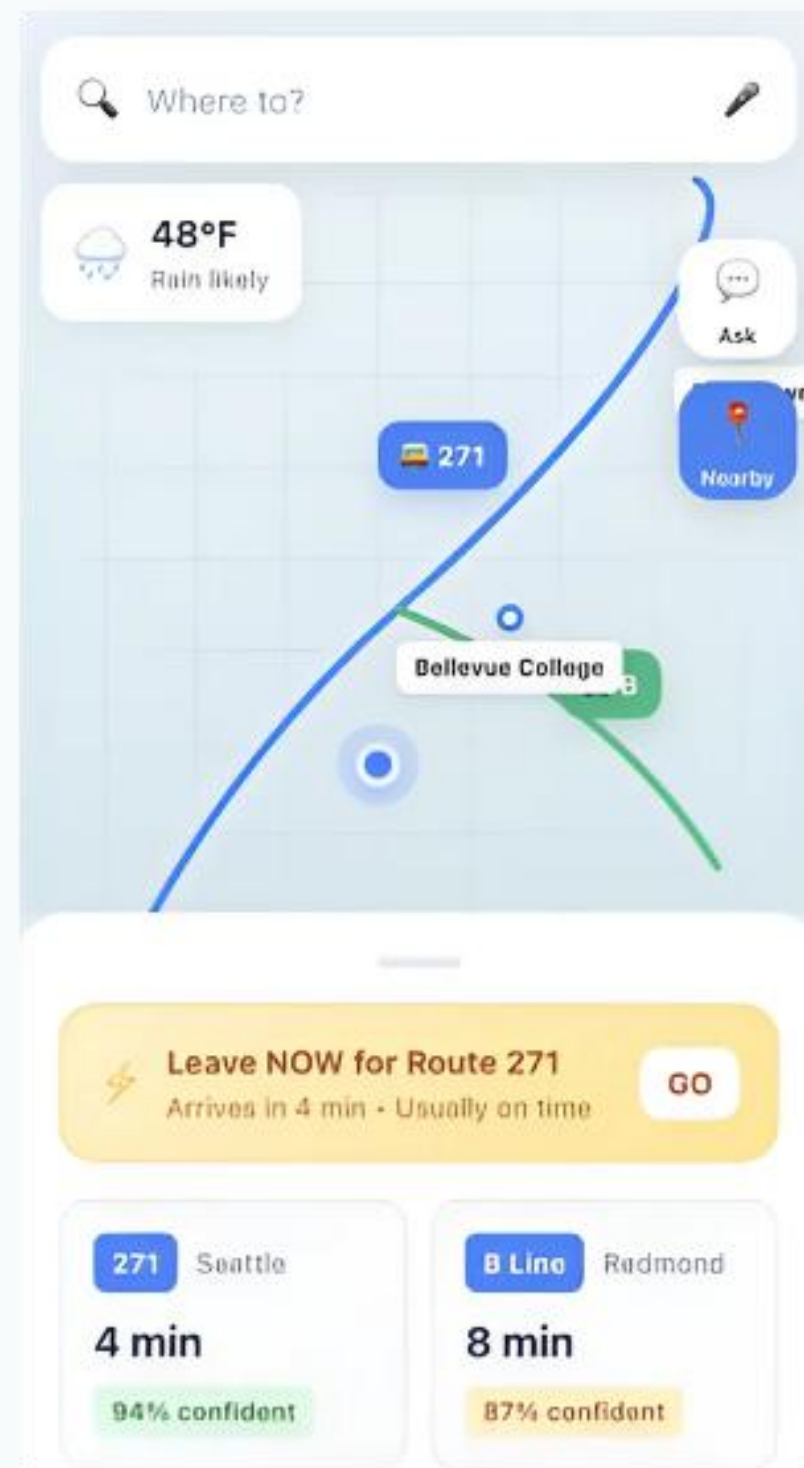
Bus tracking

Alternative route suggestions in case of delays and disaster alerts

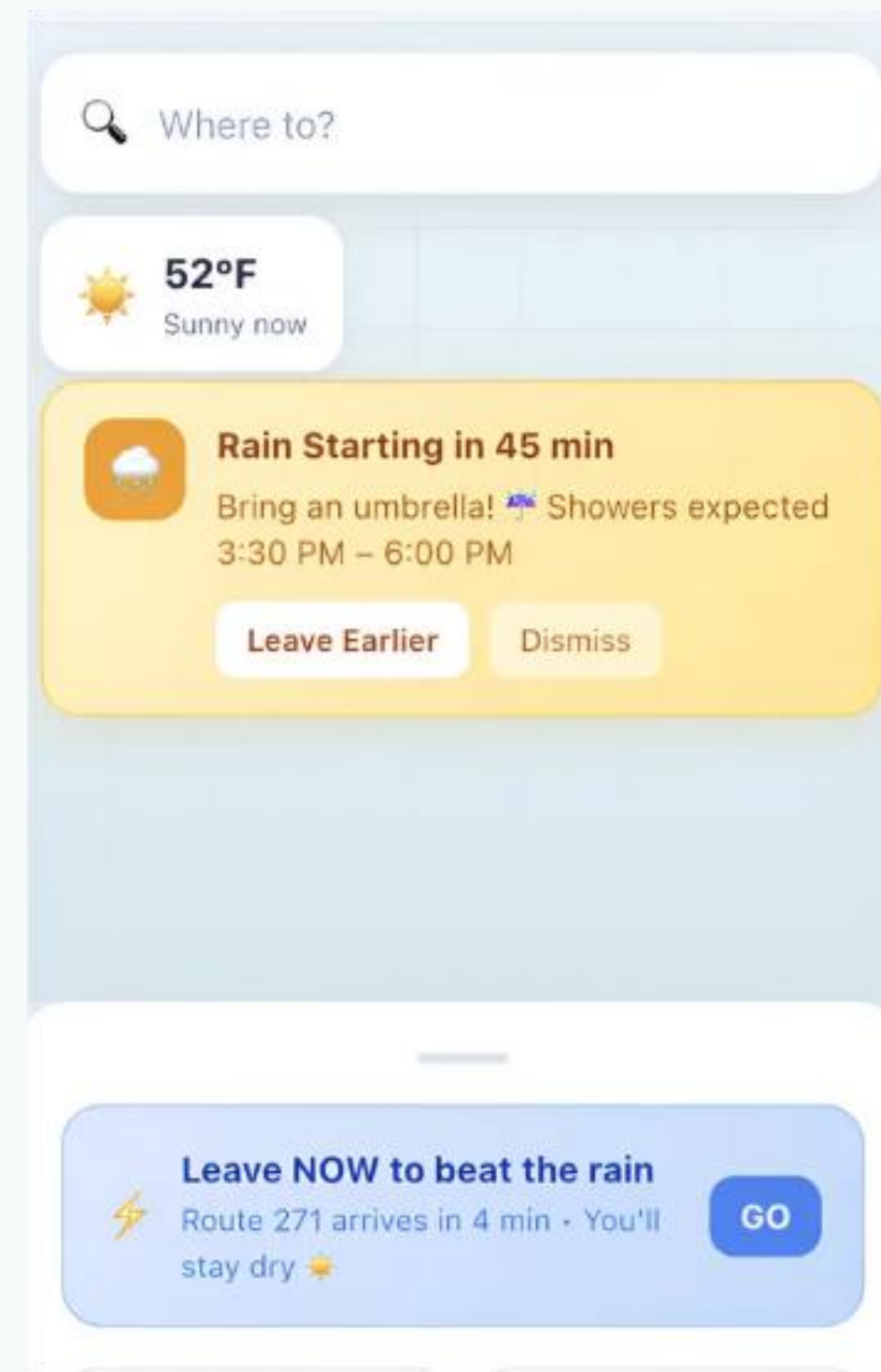


# 05. Application's Feature

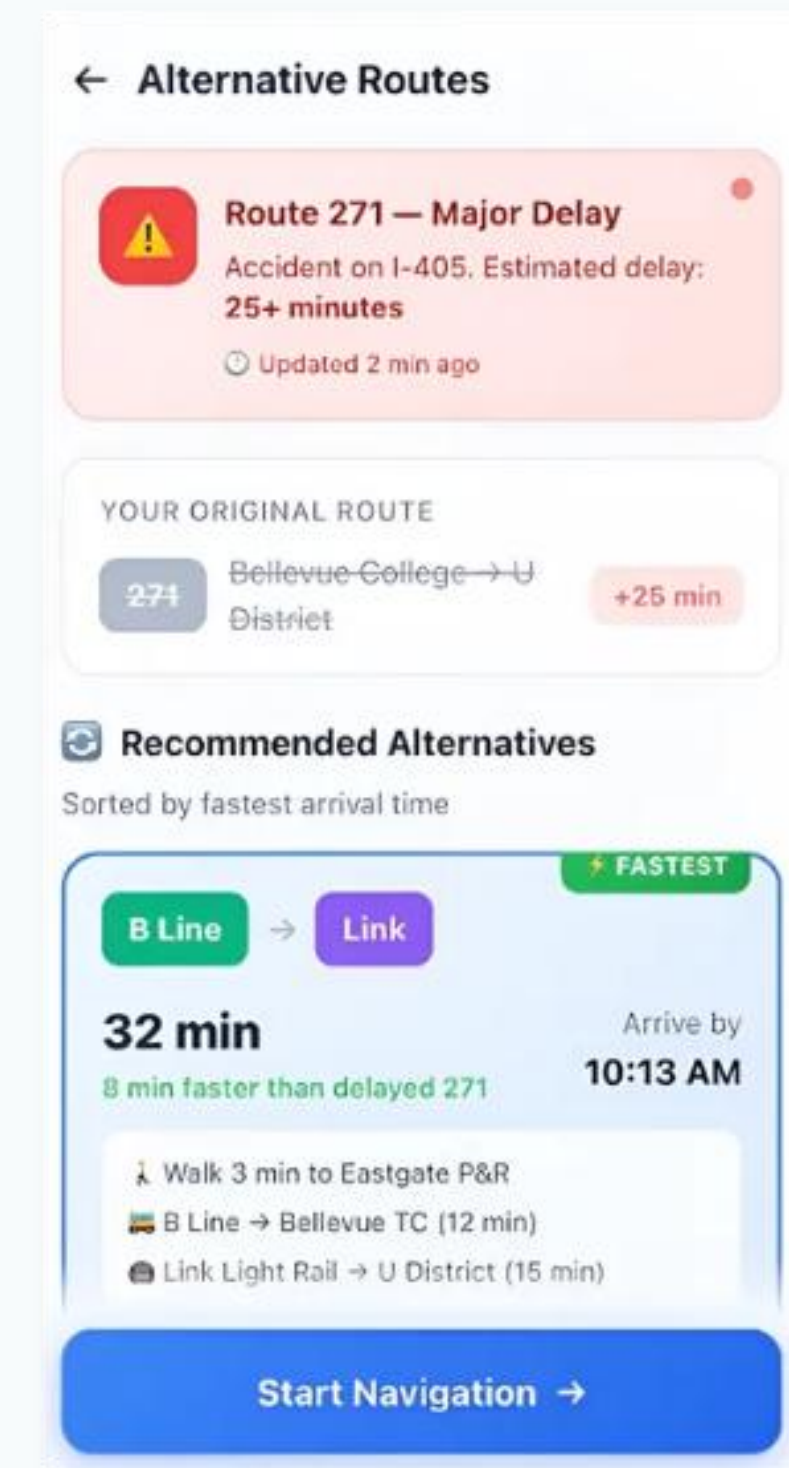
## (1) Real-Time Capability



Real-time  
bus tracking



Translated and contextualized  
weather updates



Delay-aware alternative route  
suggestions and disaster alerts



# 05. Application's Feature

## (2) Data Integration

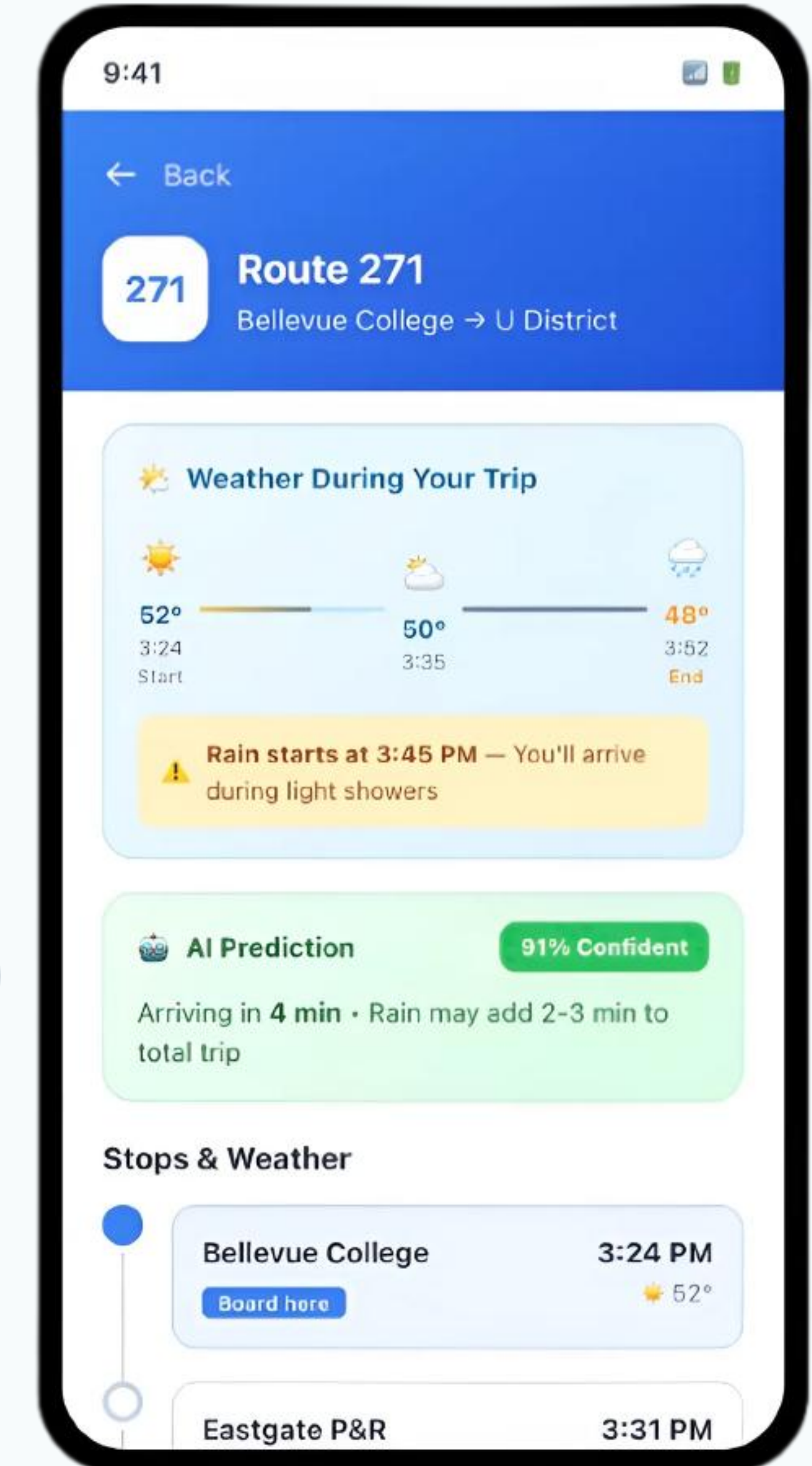
Real-time weather  
and climate impact

Real-time bus operation  
information

AI-based optimal route  
recommendation

Bus crowding and  
full-capacity status

Integrated traffic control  
and delay information



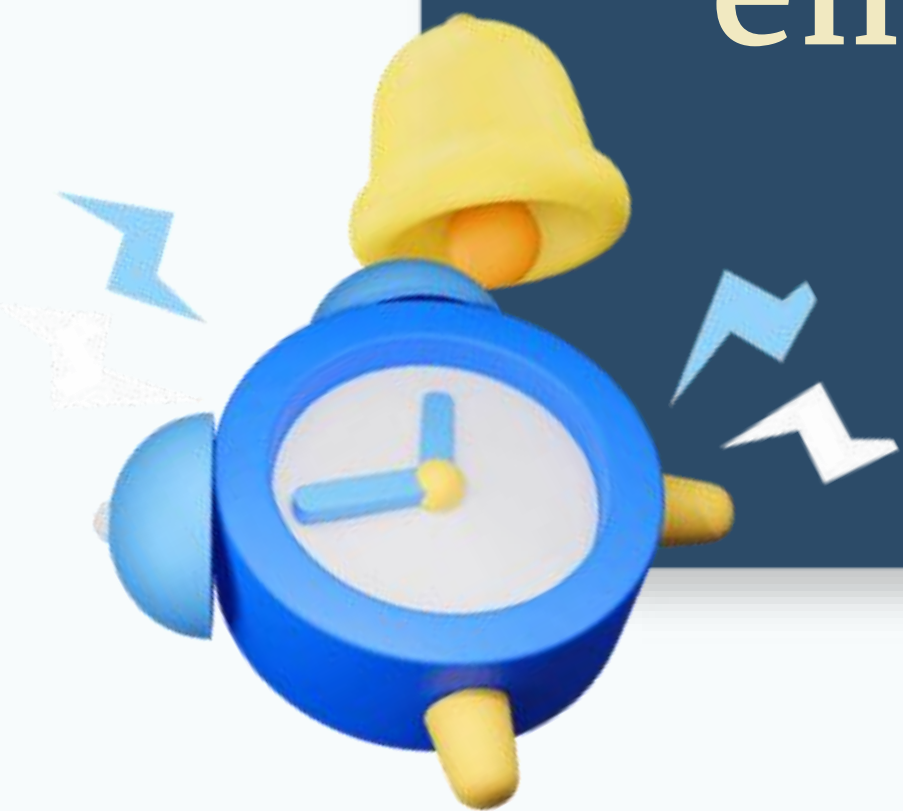


# 06. Project Process

(1) Revising MVP

**MVP**

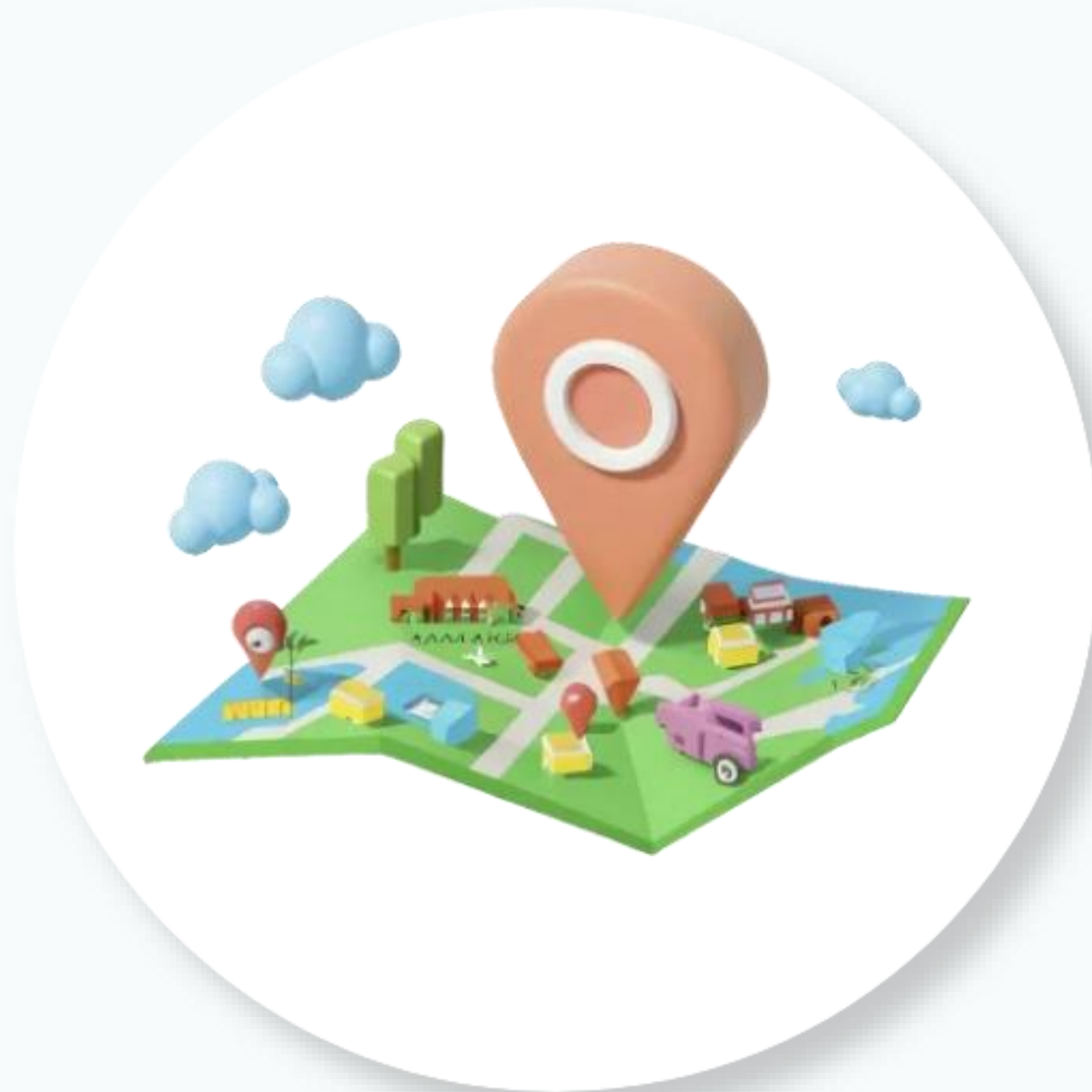
Visualize weather APIs and  
emergency alert APIs by mapping them with  
bus route GPS data in real-time.



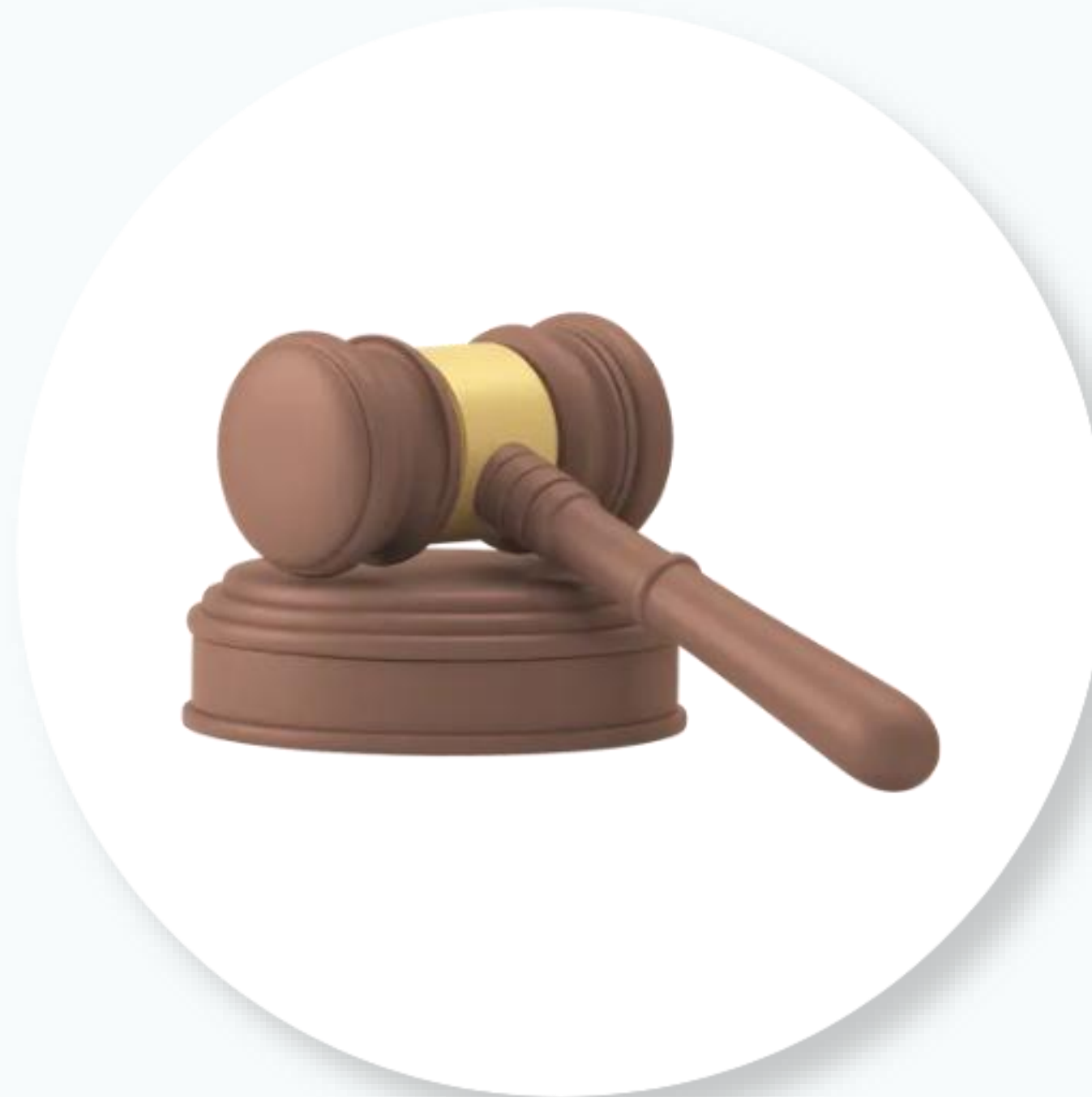


# 06. Project Process

## (2) Feedback Reflection



Need for differentiation



Privacy & Legal issue



Crowdsourcing

# 07. Next Step



## Industry Visit – DigiPen (Jan 15th)

Interviews with Korean and international students at DigiPen Institute of Technology

### Goal

Identify public transportation usage patterns among local and international students.



## Field Research (Jan 17th)

Observation of public transit users and on-site interviews



## Public Holiday Utilization (Jan 19th)

Organization and analysis of field research results



# Team Objective Progress Over Perfection

Always has been.

It is just Google Map with  
Weather and  
crowdsourcing Bus Data.





Q&A



# \* Extra Slide – Persona



## <Persona Setting>

### Target:

A working mother living in the suburbs of Seattle who relies on public transportation.

### Core Pain Point:

A user for whom transportation delays directly lead to anxiety regarding her livelihood

# \* Extra Slide – Scenario



**Morning (10:00 AM)**

Bus delays threaten her on-time arrival, causing anxiety about job security.



**Afternoon (14:00 PM)**

uncertain transit schedules create ongoing stress around childcare timing.



**Night (21:00 PM)**

OReturn-trip delays force trade-offs between work and family responsibilities.